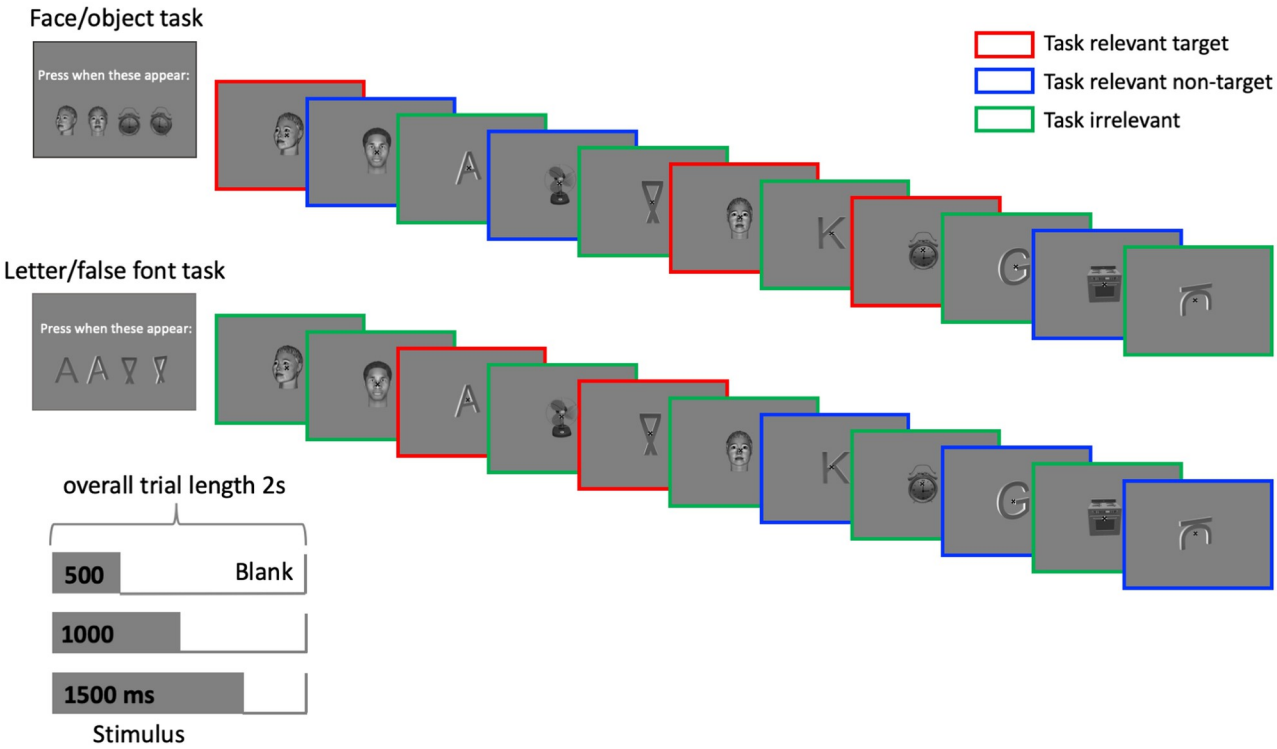

Fig 2. Experiment 1 design—In separate blocks (top & bottom rows), subjects will be asked to detect infrequent targets drawn from two categories: a specific face and a specific object (upper row), or a specific letter and a specific false-font (lower row). The stimuli will be shared across blocks, but the selection of the targets will determine to which of three trial types each stimulus belongs: *task-relevant targets*, *task-relevant non-targets*, *task-irrelevant non-targets* highlighted by red, blue and green frames respectively. Colored frames are shown for illustration purposes only. The duration (500, 1000, 1500 ms) and orientation (side view/front view) of stimuli will be manipulated to allow for specific analyses. Blank intervals between stimuli and intertrial intervals (truncated exponential distribution) are not depicted here.

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an electrographic seizure within 3-hours prior to testing. When available, information concerning language dominance as assessed by the intracarotid sodium amobarbital (Wada) will be reported.

Informed consent will be obtained prior to the study. In the case of minors, assent will be obtained from the minor and informed consent will be obtained from a parent or a legal guardian. IRB approval has been obtained from each local site in which the experiments are being conducted. Patients will be informed that participation in the study will not affect their clinical care and that they could withdraw their participation from the study at any point without affecting their clinical care. All subjects will be informed that data will be made publicly available for a minimum of 5 years after the publication of the main results. Only subjects agreeing to the data sharing procedure will be recruited.

Targeted sample sizes were determined for each methodology as being 2.5 times larger than common sample sizes in the literature for that methodology (50 subjects for fMRI and for M-EEG given that 20 subjects is the common sample size; 25 subjects for iEEG given that 10 subjects is the common sample size). Samples sizes refer to total data that will be collected in each individual laboratory, i.e., total sample sizes when the data is combined across labs will be one hundred for fMRI, one hundred for M-EEG, and fifty for iEEG. We estimate a 20% attrition related to data quality. Even after considering attrition, since we will use a within-subject design, this sample size will give us >90% power to detect differences of medium effect size (Cohen's d = >0.5).

The data will be analyzed by the data monitoring team (who are not part of the data acquisition labs) to ensure data quality. This will be done both behaviorally, aimed at excluding subjects who do not meet our predefined criteria (see below), and physiologically, to test data quality (e.g., by assessing signal to noise ratio in the neural data, eye movement patterns in the eye tracking data, etc.). These data quality checks are orthogonal to the tested predictions by the theories. We will stop collecting data upon completing the predefined sample size. No further rejection of subjects will occur after this stage.

The exclusion criteria will include: (a) insufficient number of trials in each of the experimental conditions, due to excessive eye movements, muscular artifacts, movement, noisy recording, or subjects deciding to stop the experiments (<30 for M-EEG or <20 for fMRI. If analyses show that a good enough signal can be obtained with fewer trials, these numbers will be amended); and (b) low performance in the attention tasks. In Experiment 1, this translates into: <80% hits, >20% FAs for fMRI and M-EEG subjects; <70% hits, >30% FAs for iEEG patients. For Experiment 2 performance criteria, see below.

Overview of operational hypotheses of the theories

Based on the opposing predictions of the theories, we will focus on testing three predictions. The first refers to **decodability of content of consciousness** (see analysis #1 below). Decoding will be tested for category as well as for orientation (which is a task irrelevant feature, because–as opposed to category–it is completely orthogonal to the task). As the stimuli are clearly visible in this experiment, the theories predict that information about the content of consciousness should be consistently found in the underlying neural substrates, even if the stimulus (or one of its features, like orientation) is task-irrelevant. Thus, according to GNW, such information should be found in nodes of prefrontal areas, while according to IIT it should be primarily found in posterior areas. The critical test of these predictions will accordingly be whether decoding of category and orientation generalizes from the task relevant (non-target) condition to the task irrelevant condition, within the predicted brain regions (PFC for GNW versus posterior hot zone for IIT). The second prediction refers to the **maintenance of a percept over time**, manipulated here via stimuli duration (see analysis #2 below). GNW predicts transient activity, reflecting an updating mechanism at stimulus onset and offset, while IIT predicts that the physical substrate of consciousness should remain in a similar activity pattern throughout the duration of the conscious experience (here, as long as the stimulus was presented). The third prediction focuses on **inter-areal communication** between different cortical regions during conscious perception (see analysis #3 below). GNW and IIT postulate different nodes (topologies) related to consciously perceiving a stimulus: for GNW, long-range synchrony is predicted between prefrontal and category-selective areas in high-level sensory cortex; for IIT synchronization is predicted within the posterior hot-zone, between category-selective regions in high-level sensory cortex and lower-level visual areas. Finally, an additional goal of this experiment is to **pinpoint the putative NCC**, while detecting areas that cannot be regarded as such, as their activity reflects task related processes (see analysis #4 below). This holds implications for the theories, as they differ in their predictions about the NCC (see'*Neural Activations related to consciousness*' above).

Planned analyses and expected outcomes

1. Multivariate decoding. To test which cortical areas contain information about clearly perceived stimuli across both task-relevant and task-irrelevant conditions, we will decode stimulus category (face/object or letter/false-font) and orientation (side view/front view) in the task-relevant non-target condition and then test for cross-task generalization of decoding to

the task-irrelevant condition. Cross-task generalization from task-irrelevant to task-relevant condition will also be tested. Category decoding will always be carried-out within each stimulus class (faces vs. objects and letters vs. false-fonts), rather than between classes which would confound category and task (i.e., we will not decode faces vs. letters, or objects vs. false-fonts). Note that as opposed to information on category, which will sometimes be relevant and sometimes irrelevant, information on orientation will always be task-irrelevant during the entire experiment, despite being clearly seen. As mentioned above, here and in all analyses, we will use the first half of the data to optimize our analyses during phase 2 of the project (analysis development stage), including the selection of optimal decoding schemes (options include: linear SVMs, Bayesian SVMs, LDA, naïve Bayes, K-nearest neighbor, logistic regression, random forest).

GNW predicts that conscious content will be decodable from areas in the prefrontal cortex i.e., inferior frontal gyrus, middle frontal gyrus, and cingulate cortex [21, 51, 52], instantiating the global workspace, both for task relevant and task irrelevant conditions. Thus, GNW would expect transient information about category and orientation to be decodable from the PFC and global-workspace recipient regions in posterior and higher-order sensory cortex in task-relevant conditions, and for decoding to generalize between the task-relevant and task-irrelevant conditions and vice versa, specifically during the 300–500 ms time window. IIT predicts that content-specific NCCs will primarily be found in the posterior hot zone, independent of task-relevance, and makes no specific predictions as to the relevant time window. Thus, decoding of categories and orientations and cross-task generalization of decoding between the task-relevant and task-irrelevant conditions and vice versa should be maximal in posterior cortical areas.

2. Activation patterns across stimulus durations. This analysis will target the relation between the temporal duration of the experience and the neural responses. Since subjects will observe clearly visible isolated stimuli at fixation and without competition, we assume that they will experience the stimuli for as long as they are on-screen, in particular because subjects' fixation (via eye-tracking) will be constantly monitored (see the OSF page for a control experiment evaluating visibility of long-duration task irrelevant stimuli).

Here, we will evaluate the neural responses associated with the maintenance of a percept in consciousness. Specifically, we will investigate the temporal profile of activation evoked by stimuli of different durations as well as the temporal profile of informational content of the neural responses throughout the duration of the stimulus. For the former, we will compare activation levels between conditions (defined by stimulus duration) in four different time windows: (1) 300–500 ms, (2) 800–100 ms, (3) 1300–1500 ms, and (4) 1800–2000 ms using linear mixed models. For the latter, we will combine temporal generalization [53] and representational similarity analysis [RSA; 48] approaches: in long duration trials, stimulus category (as well as orientation) will be decoded in a temporal generalization fashion (testing for generalization of neural patterns containing the content specific information across time). Representational similarity analysis will then be employed to compare the resulting temporal generalization matrix to the predictions of the theories in the specified time windows.

GNW predicts ignitions within PFC following stimulus onset and possibly also during stimulus offset, constituting updates to conscious perception, with virtually no activity between updates [except for occasional stochastic bursts of spontaneous reactivation; 54]. Thus, for consciously perceived but task-irrelevant stimuli, neural activation is predicted to reflect an interaction between duration and time window in the PFC, such that activation is maximal in windows following stimulus onset and offset and minimal elsewhere. GNW further predicts that the patterns of activation will be consistent and content-specific between the windows where activation is expected to be maximum, whereas in the windows of minimal activation,